

Chapter 4 — Convection-dominated problems

Where the Galerkin method fails, and why

MTH3340 · Numerical Methods for PDEs

1 A new kind of problem

So far, everything was **elliptic**:

$$\int_{\Omega} \nabla u \cdot \nabla v + \int_{\Omega} uv = \int_{\Omega} fv \quad \text{symmetric, coercive} \implies u \in H^1(\Omega).$$

Lax–Milgram, Céa, interpolation: everything worked. Elasticity too.

Now add a term that **transports** u along a velocity field.

- the problem becomes **mixed elliptic–hyperbolic**;
- when transport dominates, Galerkin produces **large spurious oscillations**.

This week: *why*.

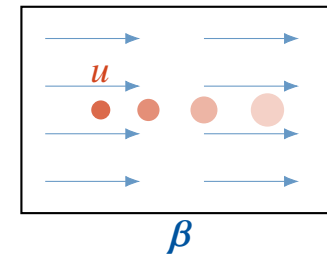
Next week: the fix.

2 The convection–diffusion–reaction equation

$$\underbrace{-\nabla \cdot \mu \nabla u}_{\text{diffusion}} + \underbrace{\sigma u}_{\text{reaction}} + \underbrace{\nabla \cdot (\beta u)}_{\text{convection}} = f \quad \text{in } \Omega, \quad \begin{aligned} u &= g \quad \text{on } \Gamma_D, \\ \mathbf{n} \cdot \mu \nabla u &= q \quad \text{on } \Gamma_N. \end{aligned}$$

$u(\mathbf{x})$: concentration of a contaminant in a fluid of velocity β .

- $\mu > 0$: diffusion (molecular mixing);
- $\sigma \geq 0$: reaction (chemical decay);
- β : the flow carries u along;
- f : sources and sinks.

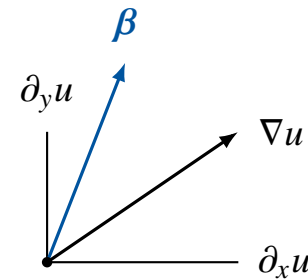


Which term dominates? Large μ : elliptic, Chapters 1–3 apply. $\mu \rightarrow 0$: the equation changes nature.

Reading the convective term

$$\nabla \cdot (\beta u) = u \underbrace{\nabla \cdot \beta}_{=0 \text{ (incompressible)}} + \beta \cdot \nabla u \quad \Longrightarrow \quad (\beta \cdot \nabla)u = \sum_{i=1}^d \beta_i \partial_i u.$$

- $\nabla u \in \mathbb{R}^d$, but $(\beta \cdot \nabla)u \in \mathbb{R}$: the **directional derivative** $\nabla_{\beta} u$;
- how u changes *moving with the flow*; blind to variations across it;
- **first order, not symmetric**. Both will matter.



3 The hyperbolic limit

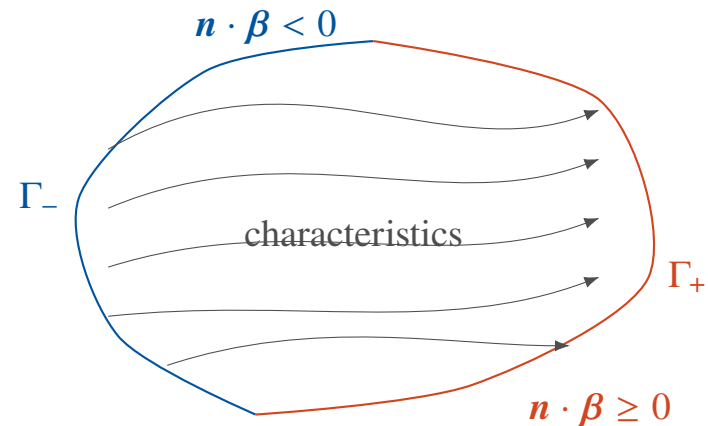
$$\mu = 0, \nabla \cdot \boldsymbol{\beta} = 0:$$

$$(\boldsymbol{\beta} \cdot \nabla)u + \sigma u = f \quad \text{in } \Omega. \quad \text{First order.} \quad \text{Where can we prescribe } u?$$

$$\Gamma_- \doteq \{x \in \partial\Omega : \boldsymbol{n} \cdot \boldsymbol{\beta} < 0\} \quad \text{inflow},$$

$$\Gamma_+ \doteq \partial\Omega \setminus \Gamma_- \quad \text{outflow}.$$

- Dirichlet data **only on the inflow**:
 $u = g$ on Γ_- , $\Gamma_D \subset \Gamma_-$;
- what arrives at Γ_+ is *determined* by the equation.



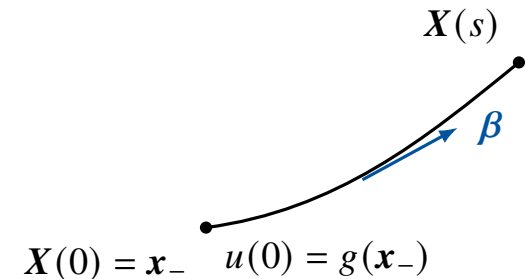
Characteristics

Fix $\mathbf{x}_- \in \Gamma_-$ and follow the flow:

$$X(0) = \mathbf{x}_-, \quad \frac{dX}{ds} = \beta(X(s)).$$

Chain rule along the curve:

$$\frac{d}{ds} u(X(s)) = \nabla u \cdot \frac{dX}{ds} = \beta \cdot \nabla u = f - \sigma u.$$



The PDE is an ODE along each characteristic, with initial value $g(\mathbf{x}_-)$.

- each characteristic is solved **independently**: information travels along β , never across;
- a jump in g is transported: u is discontinuous across characteristics. No $H^1(\Omega)$ here.

What diffusion changes

Put back $-\nabla \cdot \mu \nabla u$, $\mu > 0$: elliptic again, and everything is overturned.

hyperbolic, $\mu = 0$

discontinuities across characteristics

Dirichlet data **only on Γ_-**

diffusive, $\mu > 0$

$u \in H^1(\Omega)$; a jump in g smeared over a layer of width $O(\mu^{1/2})$

Dirichlet data **also on Γ_+** ; if incompatible, a **boundary layer** of width $O(\mu)$

$\mu \rightarrow 0$ is a **singularly perturbed** limit: the type of the PDE changes, and the layers become thinner than any mesh.

Example: a step carried across the square

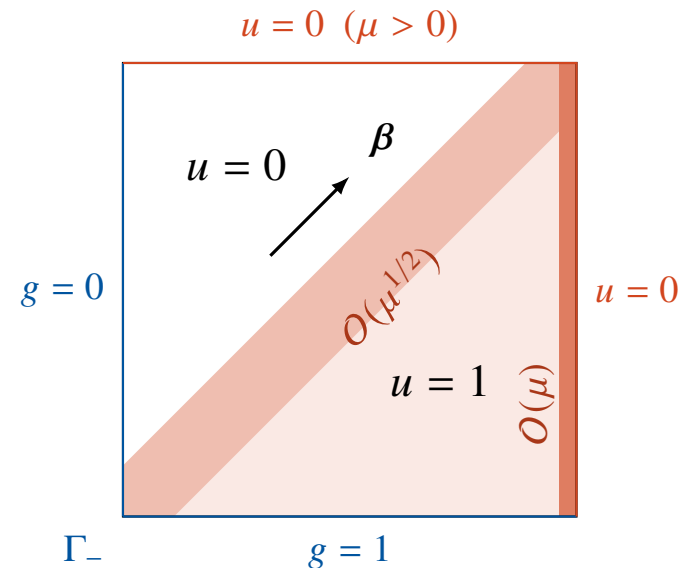
$\Omega = (0, 1)^2$, $\beta = (1, 1)/\sqrt{2}$, $f = 0$,
 $\sigma = 0$.

Inflow data: $g = 1$ (bottom), $g = 0$ (left).

$\mu = 0$: $u = 1$ below the diagonal, $u = 0$ above. The corner jump is transported intact.

$\mu > 0$ small, with $u = 0$ also on the outflow:

- **internal layer** of width $O(\mu^{1/2})$ along the diagonal;
- **boundary layer** of width $O(\mu)$ on the right edge.



4 The weak form

Assume $g = 0$, $\Gamma_D = \partial\Omega$, constant $\mu > 0$, $\sigma \geq 0$, $\nabla \cdot \boldsymbol{\beta} = 0$. Multiply by $v \in H_0^1(\Omega)$ and integrate by parts *only the diffusive term*.

Definition (Weak convection–diffusion–reaction problem)

Find $u \in H_0^1(\Omega)$ such that, for all $v \in H_0^1(\Omega)$,

$$a(u, v) \doteq \int_{\Omega} \mu \nabla u \cdot \nabla v + \int_{\Omega} v (\sigma u + (\boldsymbol{\beta} \cdot \nabla) u) = \int_{\Omega} f v \doteq \ell(v).$$

- diffusion and reaction: the terms we know;
- convection $\int_{\Omega} v (\boldsymbol{\beta} \cdot \nabla) u$: **not symmetric** in (u, v) .

Q: Is $a(\cdot, \cdot)$ coercive? With which constant? Everything hinges on $v = u$.

The convective term is skew-symmetric

$v = u \in H_0^1(\Omega)$, and $u \partial_i u = \frac{1}{2} \partial_i (u^2)$:

$$\int_{\Omega} u (\beta \cdot \nabla) u = \frac{1}{2} \int_{\Omega} \beta \cdot \nabla (u^2) = -\frac{1}{2} \int_{\Omega} \underbrace{(\nabla \cdot \beta)}_{=0} u^2 + \frac{1}{2} \int_{\partial\Omega} (\mathbf{n} \cdot \beta) \underbrace{u^2}_{=0} = 0.$$

$$a(u, u) = \mu |u|_{H^1(\Omega)}^2 + \sigma \|u\|_{L^2(\Omega)}^2.$$

- convection contributes **nothing** to $a(u, u)$;
- what is left is the coercivity of **diffusion and reaction alone**.

Convection is invisible to the energy $a(u, u)$, yet it can be the largest term in the equation.

The two constants

Continuity. Cauchy–Schwarz term by term, with $|(\boldsymbol{\beta} \cdot \nabla)u| \leq \|\boldsymbol{\beta}\|_{L^\infty} |\nabla u|$ pointwise:

$$\begin{aligned} |a(u, v)| &\leq \underbrace{\mu |u|_{H^1} |v|_{H^1}}_{\text{diffusion}} + \underbrace{\sigma \|u\|_{L^2} \|v\|_{L^2}}_{\text{reaction}} + \underbrace{\|\boldsymbol{\beta}\|_{L^\infty} |u|_{H^1} \|v\|_{L^2}}_{\text{convection}} \\ &\leq M \|u\|_{H^1} \|v\|_{H^1}, \quad M \doteq \mu + \sigma + \|\boldsymbol{\beta}\|_{L^\infty(\Omega)}. \end{aligned}$$

Coercivity. Convection gave nothing, so only Poincaré is left, $\|u\|_{L^2} \leq C_P |u|_{H^1}$ in $H_0^1(\Omega)$:

$$a(u, u) = \mu |u|_{H^1}^2 + \sigma \|u\|_{L^2}^2 \geq \mu |u|_{H^1}^2 \geq \alpha \|u\|_{H^1}^2, \quad \alpha \doteq \frac{\mu}{1 + C_P^2}.$$

$\boldsymbol{\beta}$ enters M and never α . That gap is the whole chapter. $M < \infty$ and $\alpha > 0$ still give Lax–Milgram: a unique solution for every $\mu > 0$.

5 Stability of the continuous problem

Nothing new to prove: take $v = u$, and use the two constants.

Theorem (Stability of the convection–diffusion problem)

$$\|u\|_{H^1(\Omega)} \leq \frac{1}{\alpha} \|f\|_{H^{-1}(\Omega)} = (1 + C_P^2) \frac{\|f\|_{H^{-1}(\Omega)}}{\mu}.$$

$$\underbrace{\alpha \|u\|_{H^1}^2}_{\text{coercivity of } a} \leq a(u, u) = \ell(u) \leq \underbrace{\|f\|_{H^{-1}} \|u\|_{H^1}}_{\text{continuity of } \ell},$$

then divide by $\|u\|_{H^1}$, where $\|f\|_{H^{-1}} \doteq \sup_{v \in H_0^1(\Omega)} \ell(v) / \|v\|_{H^1}$. □

The energy argument does not see β at all, although the hyperbolic problem is well behaved along characteristics. All of the cost is in $1/\alpha = (1 + C_P^2)/\mu$.

The notes keep the terms apart, $\frac{\mu}{2} |u|_{H^1}^2 + \sigma \|u\|_{L^2}^2 \leq \frac{C_P^2}{2\mu} \|f\|_{H^{-1}}^2$: same content, finer bookkeeping.

6 The Galerkin approximation

$$u_h \in V_h \subset H_0^1(\Omega) : \quad a(u_h, v_h) = \ell(v_h) \quad \forall v_h \in V_h.$$

- Chapter 3 machinery, plus one non-symmetric element matrix $\int_K \phi^i (\beta \cdot \nabla) \phi^j$;
- same a , same ℓ , hence the **same M and α** : continuity and coercivity pass to any $V_h \subset H_0^1(\Omega)$;
- so the stability proof above holds **word for word** for u_h .

Well posed and stable for every $\mu > 0$, at the same price $1/\alpha$.

Q: But how far is u_h from u ? That is Céa's lemma, with the two constants we have just computed.

Why the ratio matters

Lemma (Céa's lemma (Chapter 2))

*a continuous with constant M , coercive with constant α , and $V_h \subset V$.
Then*

$$\|u - u_h\|_V \leq \frac{M}{\alpha} \inf_{v_h \in V_h} \|u - v_h\|_V.$$

By Galerkin orthogonality, $\alpha\|u - u_h\|^2 \leq a(u - u_h, u - u_h) = a(u - u_h, u - v_h) \leq M\|u - u_h\| \|u - v_h\|$.

$$\frac{M}{\alpha} = (1 + C_P^2) \frac{\mu + \sigma + \|\beta\|_{L^\infty}}{\mu} \sim C_\Omega \frac{\|\beta\|_{L^\infty}}{\mu} \doteq \text{Pe} \xrightarrow{\mu \rightarrow 0} \infty.$$

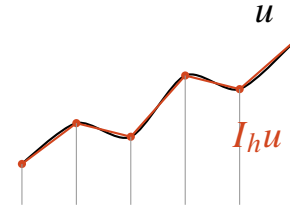
Pe is the **global Péclet number**. V_h may approximate u superbly and u_h still be Pe times worse than the best approximation.

Q: Is Céa pessimistic, or is the method really this bad?

Error analysis, step 1: Galerkin orthogonality

Split through the interpolant $I_h u \in V_h$:

$$e \doteq u - u_h = \underbrace{(u - I_h u)}_{e_{Ih}: \text{interpolation}} + \underbrace{(I_h u - u_h)}_{e_h: \text{discrete}, \in V_h}.$$



$a(u, v_h) = \ell(v_h) = a(u_h, v_h)$ for all $v_h \in V_h$, hence

$$a(u - u_h, v_h) = 0 \quad \implies \quad a(e_h, v_h) = -a(e_{Ih}, v_h).$$

Take $v_h = e_h$ (allowed: $e_h \in V_h$). Convection drops out on the left:

$$\mu |e_h|_{H^1(\Omega)}^2 + \sigma \|e_h\|_{L^2(\Omega)}^2 = -a(e_{Ih}, e_h).$$

Error analysis, step 2: the three terms

$$-a(e_{Ih}, e_h) \leq \left| \int_{\Omega} \mu \nabla e_{Ih} \cdot \nabla e_h \right| + \left| \int_{\Omega} \sigma e_{Ih} e_h \right| + \left| \int_{\Omega} e_h (\boldsymbol{\beta} \cdot \nabla) e_{Ih} \right|.$$

Diffusion, reaction: Young ($ab \leq a^2 + \frac{1}{4}b^2$), giving $\mu|e_{Ih}|_{H^1}^2 + \sigma\|e_{Ih}\|_{L^2}^2$ plus a quarter of the left-hand side.

Convection: the left only pays for $\mu^{1/2}|e_h|_{H^1}$, so move the derivative onto e_h (parts, $\nabla \cdot \boldsymbol{\beta} = 0$, zero traces):

$$\begin{aligned} \int_{\Omega} e_h (\boldsymbol{\beta} \cdot \nabla) e_{Ih} &= - \int_{\Omega} e_{Ih} (\boldsymbol{\beta} \cdot \nabla) e_h \leq \|\boldsymbol{\beta}\|_{L^\infty} \|e_{Ih}\|_{L^2} |e_h|_{H^1} \\ &\leq \frac{\|\boldsymbol{\beta}\|_{L^\infty}^2}{\mu} \|e_{Ih}\|_{L^2}^2 + \frac{\mu}{4} |e_h|_{H^1}^2. \end{aligned}$$

One extra power of h (the L^2 norm), at the price of $1/\mu$:

$$\frac{\mu}{2} |e_h|_{H^1}^2 + \frac{\sigma}{2} \|e_h\|_{L^2}^2 \leq \mu |e_{Ih}|_{H^1}^2 + \sigma \|e_{Ih}\|_{L^2}^2 + \frac{\|\boldsymbol{\beta}\|_{L^\infty}^2}{\mu} \|e_{Ih}\|_{L^2}^2.$$

7 The error estimate

Interpolation, $|e_{Ih}|_{H^1} \leq Ch^{r-1}|u|_{H^r}$ and $\|e_{Ih}\|_{L^2} \leq Ch^r|u|_{H^r}$, then the triangle inequality. For $\sigma = 0$:

Theorem (Error estimate for convection–diffusion)

$$\mu |u - u_h|_{H^1(\Omega)}^2 \leq C \left(1 + \frac{\|\beta\|_{L^\infty(\Omega)}^2 h^2}{\mu^2} \right) \mu h^{2(r-1)} |u|_{H^r(\Omega)}^2.$$

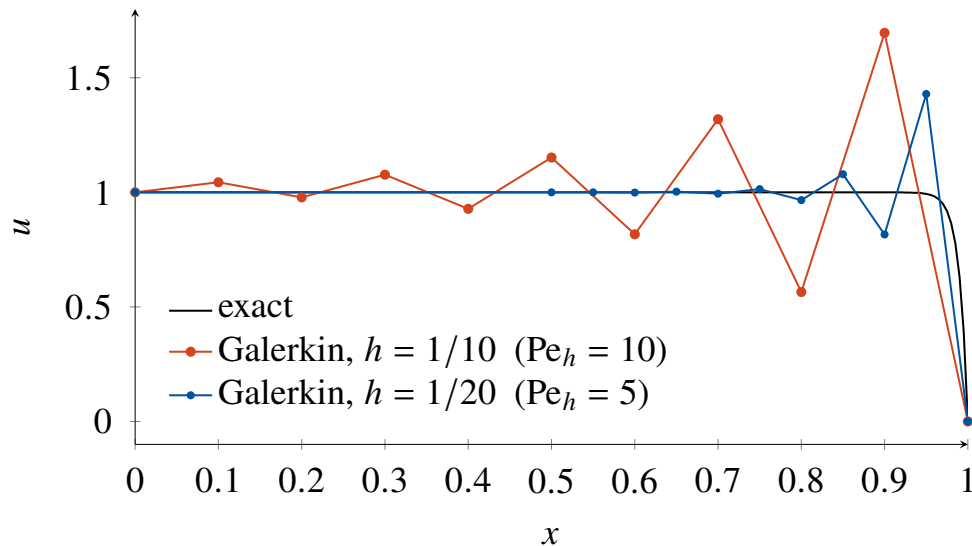
mesh Péclet number $\text{Pe}_h \doteq \frac{\|\beta\|_{L^\infty} h}{\mu} : \quad |u - u_h|_{H^1} \lesssim (1 + \text{Pe}_h) h^{r-1} |u|_{H^r}.$

- optimal *rate*; the constant is the **mesh** Péclet number $\text{Pe}_h = \text{Pe } h/L$, one power of h better than Céa's global one;
- the blow-up as $\mu \rightarrow 0$ is **real**, not an artefact of the analysis;
- cured only for $\text{Pe}_h \lesssim 1$, i.e. $h \lesssim \mu/\|\beta\|$. With $\mu = 10^{-6}$: 10^{18} cells in 3D.

Seeing it in 1D

$$-\mu u'' + u' = 0 \text{ in } (0, 1), \quad u(0) = 1, \quad u(1) = 0, \quad \mu = 10^{-2}, \quad \mathcal{P}_1.$$

- exact: $u \equiv 1$, then a layer of width $O(\mu)$ at $x = 1$;
- Galerkin **oscillates wildly** around the layer;
- halving h halves the amplitude, does not remove it;
- gone only for $\text{Pe}_h \lesssim 1$:
 $h \lesssim 10^{-2}$.



Q: Write the equation of an interior node i in terms of U^{i-1}, U^i, U^{i+1} . What happens as $\mu \rightarrow 0$? (Tutorial.)

8 Where do we go from here?

Diagnosis.

- coercivity from μ alone: **convection is skew-symmetric**, the energy cannot see it;
- error constant $\sim \text{Pe}_h = \|\beta\| h / \mu$;
- $\text{Pe}_h \lesssim 1$ by refinement: unaffordable, above all in 3D.

Way out: change the discrete problem. Next week:

1. **artificial diffusion** (Von Neumann & Richtmyer, 1950);
2. **streamline diffusion**: diffusion only along β ;
3. **SUPG**: a *consistent* stabilisation with optimal rates.

Remark

Same story for Navier–Stokes at high Reynolds number (turbulence): Galerkin is useless unless the mesh resolves the *viscous* scales. Same cure.